



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 • STANDARD 6.GR.1

Area of Parallelograms

Lesson 5-1 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can find the area of a parallelogram using $\text{base} \times \text{height}$.

Language Objective: I can explain how I found the area using the words base, height, area, and formula.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Parallelogram

Base

Height

Area

Composite figure

Formula



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A quadrilateral with two pairs of parallel sides is a

2

Any side of a parallelogram used to calculate its area is the

3

The perpendicular distance between the base and the opposite side is the

4

The amount of flat space inside a two-dimensional figure is its

5

A shape made of two or more basic shapes combined is a

6 A math rule written with symbols, like $A = b \times h$, is a .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

JUSTIFY

On the grid you cut a triangle off one end of the parallelogram and slid it to the other side. What shape did you make, and why does that prove the area is base x height?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Base** — A side of the shape you use to find the area.

Base — Un lado de la figura que usas para hallar el área.

- ☐ **Height** — The straight-up distance from the base to the top.

Altura — La distancia recta hacia arriba desde la base hasta la parte de arriba.

- ☐ **Area** — How much space is inside a flat shape.

Área — Cuánto espacio hay dentro de una figura plana.

- ☐ **Parallelogram** — A four-sided shape with two pairs of parallel sides.

Paralelogramo — Una figura de cuatro lados con dos pares de lados paralelos.

- ☐ **Composite figure** — A shape made by putting two or more simple shapes together.

Figura compuesta — Una figura formada al juntar dos o más figuras simples.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

When I move the triangle, I get a ____.

Cuando muevo el triángulo, obtengo un ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: We use the height because it is perpendicular to the base.

Evidence: Students connect 'height' to the perpendicular (straight-up) distance from base to top, and recognize the slanted side is longer and would overstate the area.

Reasoning: This evidence connects the definition of parallelogram to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **When I move the triangle, I get a ____.**
Cuando muevo el triángulo, obtengo un ____.
- **Area equals base times ____, so 14 times 9 equals ____.**
El área es la base por ____, entonces 14 por 9 es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**
Lo sé porque ____.
- **So ____.**
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Parallelogram
2. **Notes 2:** Base
3. **Notes 3:** Height
4. **Notes 4:** Area
5. **Notes 5:** Composite figure
6. **Notes 6:** Formula
7. **Watch for this mistake:** *The most common mistake in Area of Parallelograms is multiplying the base by the slanted side instead of the perpendicular height. For a parallelogram with a base of 16 ft, a perpendicular height of 9 ft, and a slanted side of 11 ft, students compute $16 \times 11 = 176$ square feet instead of the correct $16 \times 9 = 144$ square feet. The height must be the straight-up (perpendicular) distance between the base and its opposite side, not the length of the slanted side, even though the slanted side is often the number that 'looks' like it belongs in the formula.*
8. **Try It 1:** A. 40 sq ft — *Area = base $\times$ height = $8 \times 5 = 40$ sq ft. The 7 ft slanted side is not used; only the perpendicular height belongs in the formula.*
9. **Try It 2:** A. Equal: both are $10 \times 6 = 60$ sq ft because area depends on base and height, not the slant — *Area = base $\times$ height. Both have base 10 ft and height 6 ft, so both areas are $10 \times 6 = 60$ sq ft. Tilting the shape changes the slanted side but not the perpendicular height, so the area stays the same.*